

HARSH RAJ

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EDUCATION

Northeastern University , Boston, MA	May 2027 (expected)
Master of Science in Computer Science	GPA: 3.6/4.0
Delhi Technological University , Delhi, India	August 2023
Bachelor of Technology in Engineering Physics	GPA: 3.4/4.0

TECHNICAL SKILLS

Programming Languages:	Python, SQL, JavaScript, C++, Java
Cloud and DevOps:	AWS, GCP, Docker, Kubernetes, Git, CI/CD
Databases:	PostgreSQL, MySQL, MongoDB, Redis, Vector Databases
Tools and Libraries:	PyTorch, Jax, Hugging Face, Ray, LangChain, Pandas, NumPy, CUDA, Triton

PROFESSIONAL EXPERIENCE

Research Assistant, Northeastern University Jan 2025 – Present

- Working with Prof. David Bau and Prof. Robert West (EPFL) on mechanistic interpretability of reasoning traces of Large Reasoning Models (LRMs)
- Built finetuning technique for LRMs to self-annotate reasoning chains, achieving 90% instruction-following

Co-Founder, Ontocord AI (Open Source) Jan 2024 – Jan 2025

- Co-founded open-source initiative focused on creating high-quality permissive datasets for LLMs
- Developed data processing pipelines with strategies for mixing, formatting, upsampling & interleaving
- Published [MixtureVitae-300BT](#) achieving 10% performance improvement over existing permissive data sets

Applied Scientist, Vijil AI Feb 2024 – Nov 2024

- Contributed to the [vijil-evaluation](#) platform for red-teaming of enterprise AI agent workflows. Integrated more than **10** benchmark datasets for vulnerability assessment
- Developed the vijil-fuzzer for adversarial text mutation. It achieved a 85% attack rate against GPT-4o

Machine Learning Engineer, Yield Inc Sept 2022 – Jan 2024

- Built synthetic data generation and finetuning infrastructure for LLM agents - [synchaev](#) and [mandrill](#)
- Contributed to [Cacti](#), a Web3 transaction chatbot that simplifies blockchain interactions through natural language

SELECTED PUBLICATIONS

• Improving Consistency in Large Language Models through Chain of Guidance 1st Author	TMLR 2025
• Defences against Reverse Preference Attacks 3rd Author	AAAI 2025
• Transfer of Adversarial Robustness from Pretraining to Downstream 2nd Author	NeurIPS 2023
• Measuring Reliability of Large Language Models through Semantic Consistency 1st Author <i>Best Paper Award</i>	NeurIPS 2022

PROJECTS

terminal-bench (600+ GitHub ★)	Language agent sandbox evaluation framework (Co-authored)
inspect-ai (1300+ GitHub ★)	Industry-standard LLM evaluation suite (Contributor)

AWARDS AND ACHIEVEMENTS

• SWISS AI Grant Recipient – Awarded 200k GPU hours and CHF 61,000 in funding	2025
• Rank 7 – Codeforces Raif ML Round 1 Competition	2021